

Marketing Campaigns Performance : A/B Testing

Business Problem:

We are a marketing agency, our primary objective is to maximize the return on investment (ROI) for our clients' advertising campaigns. Having recently executed parallel campaigns on Facebook and Google Ads, we now need to determine which platform yields superior performance across key metrics — clicks, conversions, and cost-effectiveness — to optimize resource allocation and deliver maximum value for our clients.

Research Question:

Which advertising platform is better overall - drives higher performance in terms of clicks, conversions, and cost - effectiveness?

Executive Summary

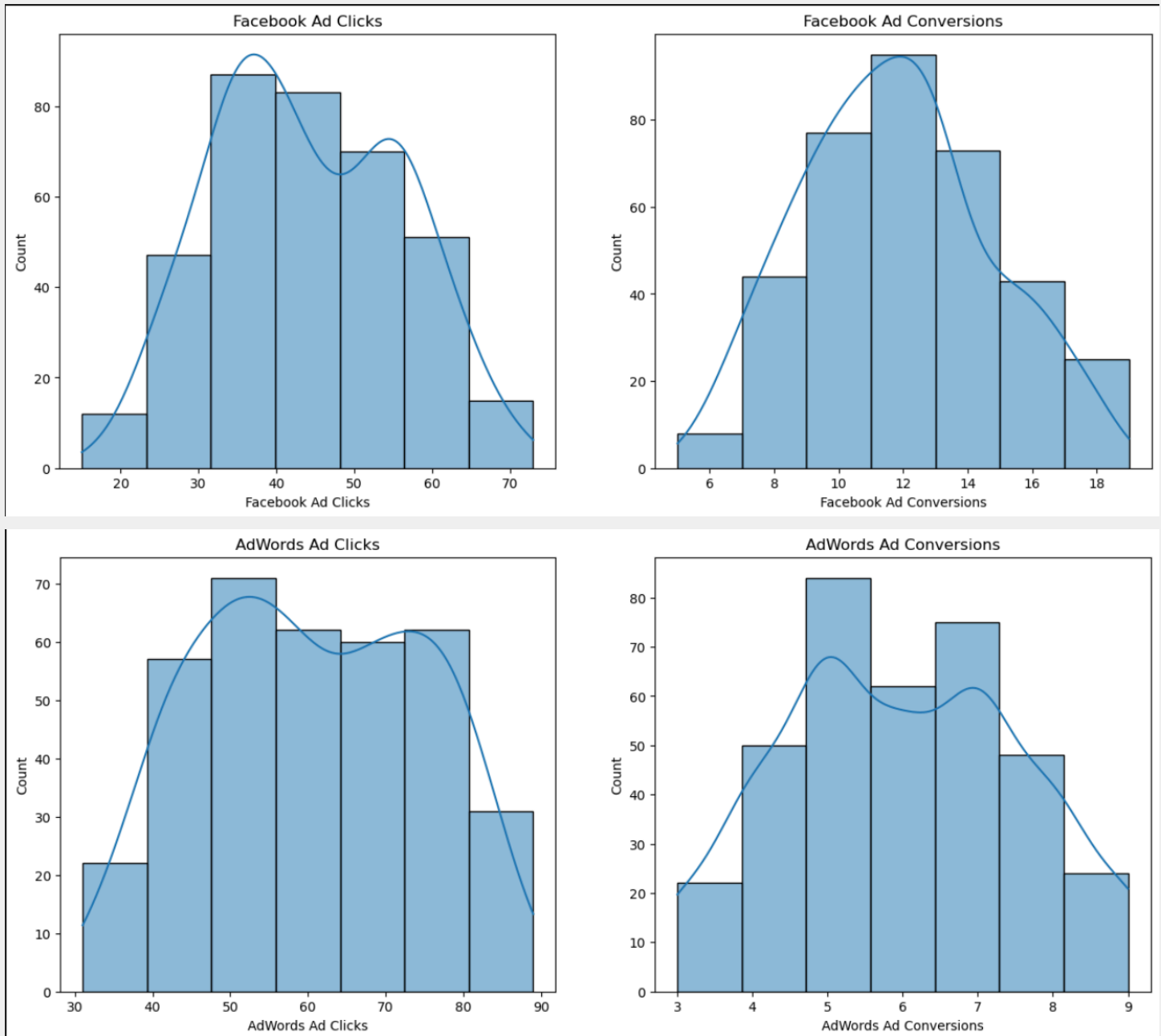
- **The Clear Platform Winner:** Facebook decisively outperformed Google AdWords across all conversion metrics. Despite taking only **39.41% (\$32K)** of total spend, Facebook generated **double** the conversion volume (**4K vs. 2K** conversions) and delivered a **5x higher overall funnel efficiency** (0.5% vs. 0.1% total conversion rate) compared to AdWords.
- **Peak Performance Windows:** Facebook acquisitions demonstrated massive Q4 efficiency, with clicks and conversions surging to annual highs in **October** (conversions >405) despite lower top-of-funnel views compared to Adwords. On a weekly baseline, **Monday** serves as the peak performance day, hitting an annual maximum of approximately 660 conversions.
- **Cost-Efficiency Benchmarks:** Facebook proved vastly superior in cost-efficiency, capturing nearly two-thirds of total acquisitions at a significantly lower baseline. While individual acquisition costs peaked in March (~\$8.73 CPC), the campaign successfully optimized in the second half of the year, hitting its most profitable benchmarks in **July (~\$6.83 CPC)** and **November (~\$6.85 CPC)**.
- **Mathematical & Statistical Confirmation:** A Welch's t-test statistically verified Facebook's massive outperformance with a T-statistic of **32.88** and an extremely low p-value of **\$9.35 \times 10^{-134}** \$, guaranteeing with mathematical certainty that these results are repeatable and not a fluke. Additionally, a cointegration test score of **-14.76** proved that Facebook spend and conversions move reliably together long-term.
- **Predictive Regression Insights:** Linear regression analysis revealed that ad clicks are a highly dependable, strong predictor of conversions (**r = 0.87**). The tightly clustered data points confirm model accuracy, allowing the agency to forecast exact outcomes with minimal risk (e.g., scaling to 80 clicks reliably yields **19.31 conversions**).
- **Core Strategic Recommendation:** Immediately invert the funding structure by **reallocating 30% to 40% of the inefficient AdWords budget directly into Facebook** to maximize client ROI, while

pivoting remaining AdWords spend strictly toward high-purchase-intent, exact-match keywords to build a unified marketing funnel.

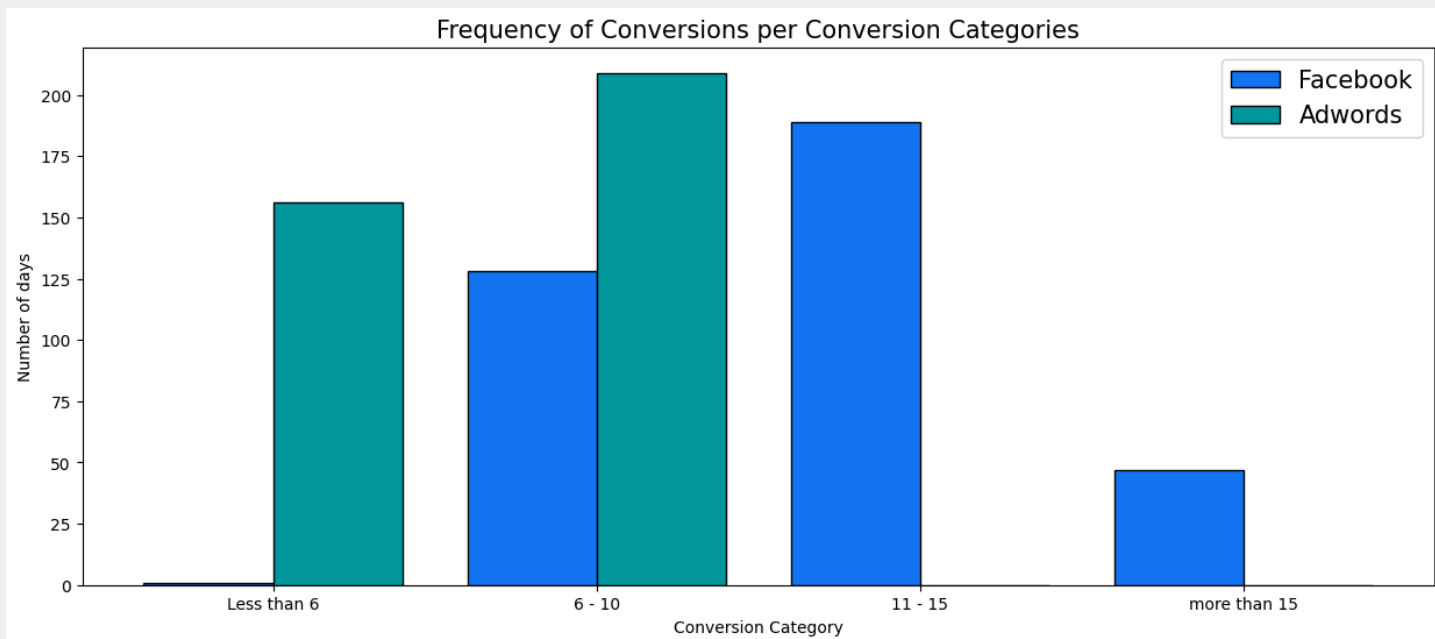
Insights Deep Dive

EDA Findings & Dashboard Insights :

Volume - Clicks & Conversions :

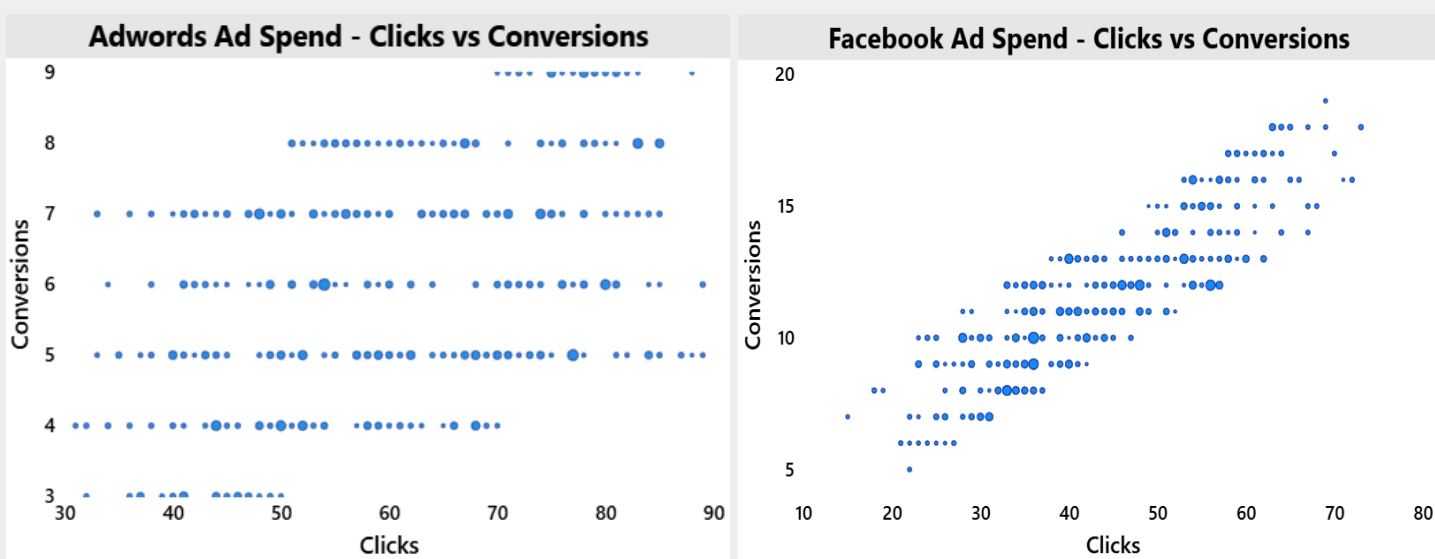


Campaign Conversions per Conversion Category by Number of Days:



- The data suggests Facebook had higher frequent conversion days than AdWords, which either had very low conversion rates (less than 6) or moderate ones (6 - 10).
- There is a significant variance in the number of high-conversion days between two different campaigns.
- The absence of any days with conversions between 10 - 15 and more than 15 in AdWords indicates a need to review what strategies were changed or what external factors could have influenced these numbers.

Ad Spend - Clicks vs Conversions:



- **Facebook Campaign:** Exhibits a highly predictable, linear trend where increased ad spend (larger bubbles) directly correlates with both higher click volume and scaling conversions.

- **AdWords Campaign:** Reflects a disjointed spend pattern. Larger bubbles (higher ad spend) are scattered arbitrarily across stratified conversion tiers (3 to 9), demonstrating that higher spend does not linearly guarantee better results.
- **Key Takeaway:** Facebook offers a clear, predictable scaling model for budget allocation, whereas AdWords spend efficiency is highly variable and non-linear.

Statistical Analysis:

Correlation Coefficient :

- **Facebook Corr. Coeff. = 0.87 :** A strong correlation ($r = 0.87$) confirms Facebook clicks are a primary driver of conversion rates. Scaling investment here offers the most direct path to growth. Facebook has high efficiency engagement which explains most conversion variance, prioritizing this channel ensures high-predictability returns.
- **Adwords Corr. Coeff. = 0.45 :** AdWords shows a moderate correlation ($r = 0.45$) with **conversions**. It contributes to results, but the relationship is less direct than Facebook. The **0.45** coefficient suggests external factors influence AdWords **conversion rates**. Further analysis is required to identify and mitigate these variables for better ROI.

Hypothesis Testing :

Hypothesis : Advertising on Facebook will result in a greater number of conversions compared to advertising on AdWords.

- Null Hypothesis (H_0): There is no difference in the number of conversions between Facebook and AdWords, or the number of conversions from AdWords is greater than or equal to those from Facebook.
- $H_0: \mu_{\text{Facebook}} \leq \mu_{\text{AdWords}}$
- Alternate Hypothesis (H_1): The number of conversions from Facebook is greater than the number of conversions from AdWords. $\mu_{\text{Facebook}} > \mu_{\text{AdWords}}$.

Findings :

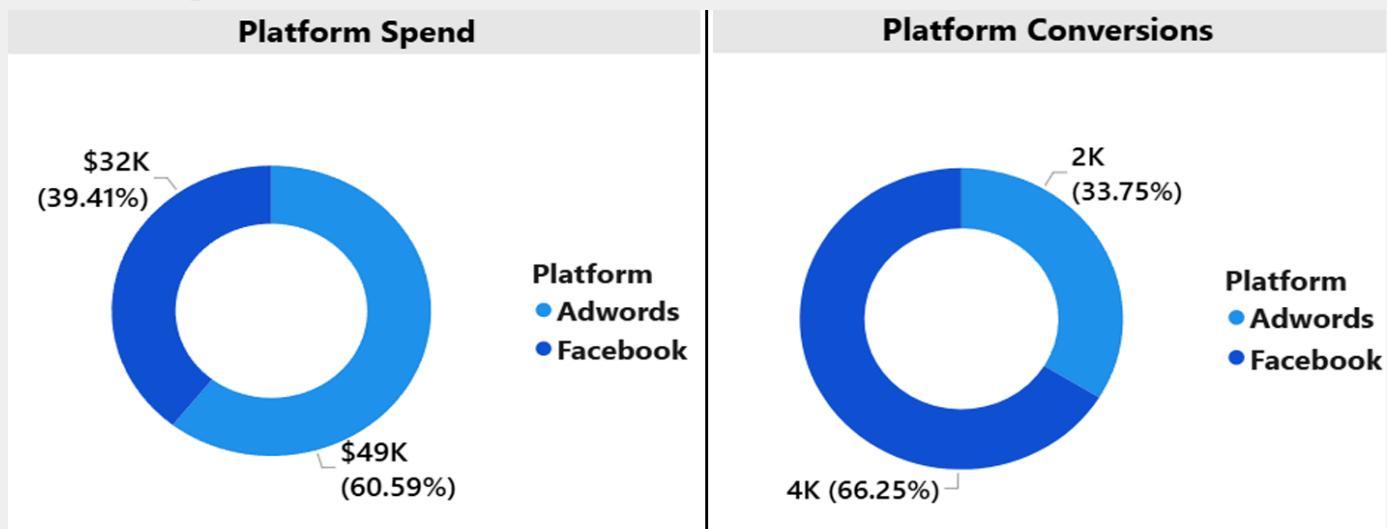
Mean Conversion	

Facebook : 11.74	T statistic 32.88402060758184
AdWords : 5.98	p-value 9.348918164530465e-134
	p-value is less than significance value, Reject the null hypothesis

- Facebook strongly outperforms AdWords in daily volume, delivering nearly double the average results with a mean conversion of **11.74 vs. 5.98**.
- Test Results: A T-statistic of **32.88** and an extremely low p-value (**$9.35 \times 10^{(-134)}$**) mean we safely reject the null hypothesis. Data strongly supports the alternate hypothesis: Facebook's conversion volume is definitively greater than AdWords.

- **The T-statistic (32.88)** measures *how big* the difference is between the two platforms. A score this high proves that Facebook's outperformance is massive, massive enough to completely rule out a close tie.
- **The P-value (9.35×10^{-134})** measures *the chance of this being a fluke*. Since it is essentially zero, it guarantees with absolute mathematical certainty that Facebook's superior performance is real and repeatable, not random luck.
- **Conclusion:** This mathematically validates aggressively shifting budget allocation away from AdWords and into Facebook to maximize client ROI.

Platform - Spend & Conversions:

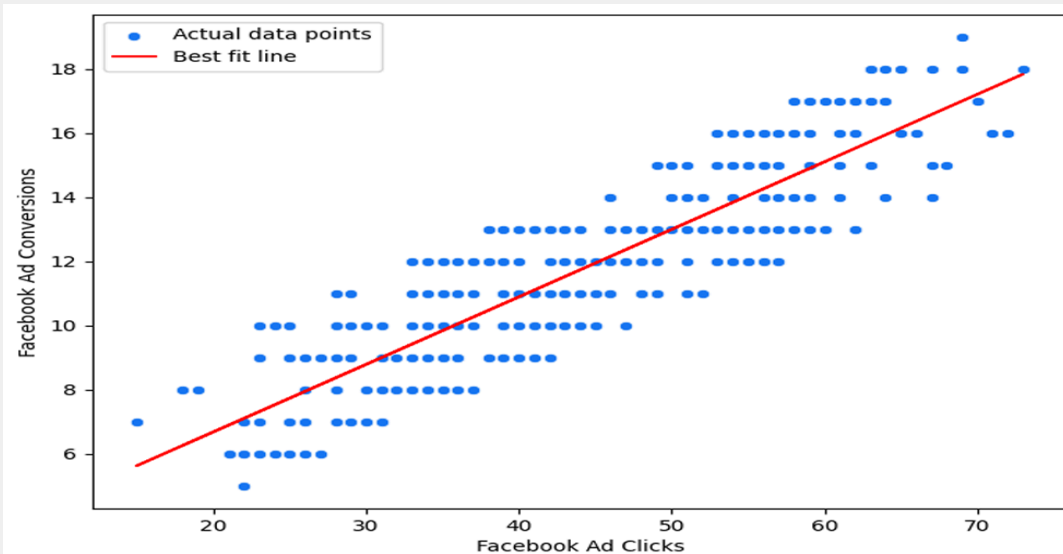


- **Resource Allocation:** Google AdWords accounted for the majority of the budget at **\$49K (60.59%)**, while Facebook received **\$32K (39.41%)**.
- **Performance Output:** Despite a significantly lower budget, Facebook delivered **4K conversions**, doubling the **2K conversions** produced by AdWords.
- **Efficiency Insight:** Facebook demonstrates vastly superior cost-efficiency, capturing nearly two-thirds of total conversions with less than 40% of the aggregate spend.

We can say Facebook is better as a Platform compared to Adwords so lets proceed with Facebook for further detailed analysis of this campaign platform.

Analysing Facebook Campaign in Detail

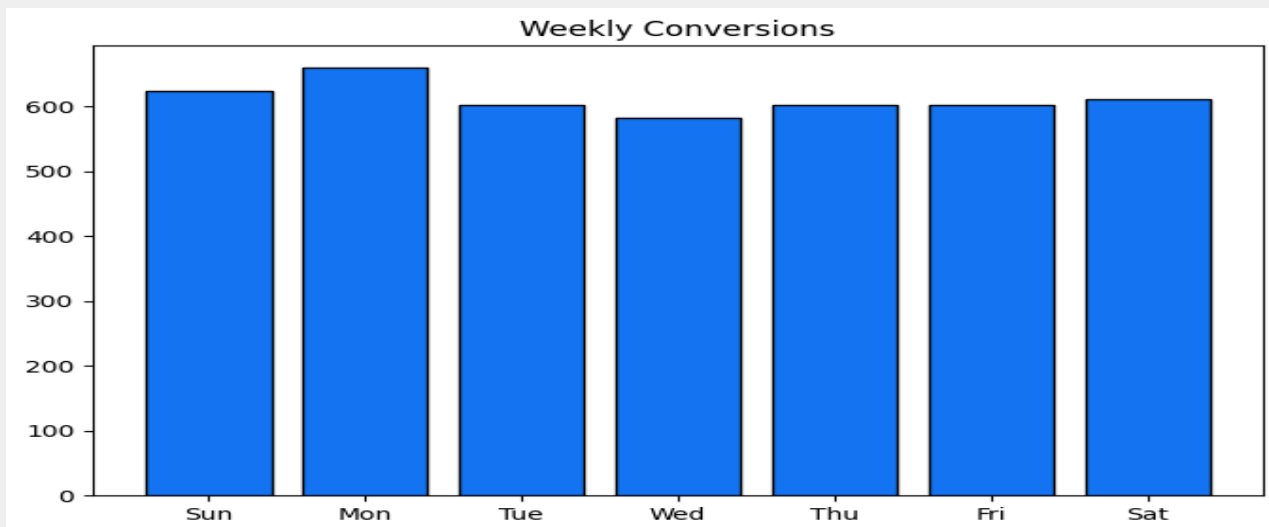
Regression Analysis



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For 50 Clicks, Expected Conversion : 13.0
For 80 Clicks, Expected Conversion : 19.31
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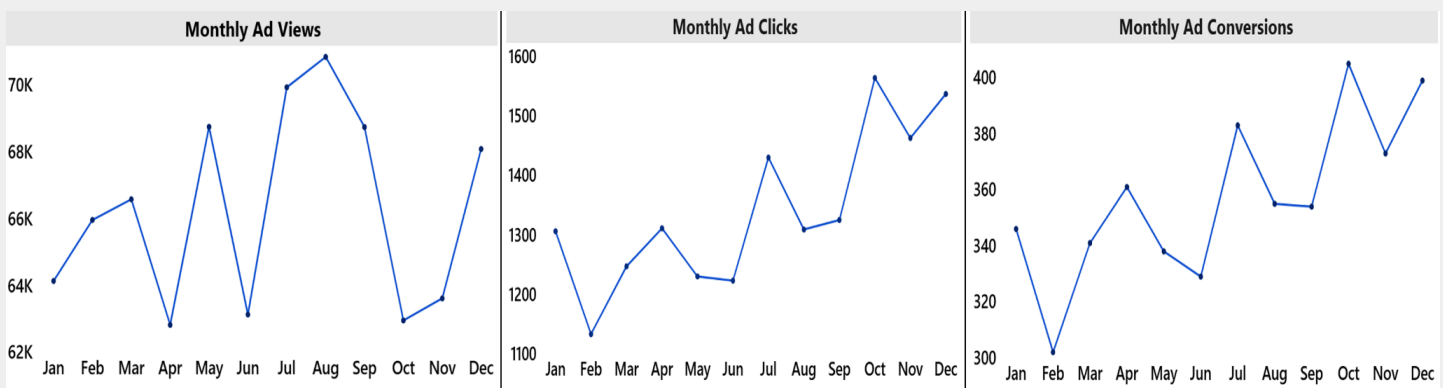
- **The Trend Line** : The red "Best fit line" tracks a highly predictable, upward pattern. As clicks increase, conversions scale up at a steady, reliable pace.
- **Model Accuracy** : The actual data points cluster tightly around this line, proving that the model is highly accurate and dependable for forecasting future performance.
- **Predictions** : This model allows us to accurately forecast revenue outcomes based on traffic goals:
 - **50 Clicks** will yield an expected **13.0 conversions**.
 - **80 Clicks** will predictably scale to **19.31 conversions**.
- ***Facebook's conversion path is highly reliable. We can confidently use this model to forecast exact client outcomes and scale ad budgets with minimal risk.***

Weekly Facebook Conversions:



- **High Stability:** Daily conversions remain remarkably steady, consistently maintaining a stable baseline of roughly 600 conversions per day.
- **Peak Performance:** Monday serves as the strongest day of the week, reaching an annual maximum of approximately 660 conversions.
- **Weekly Low:** Wednesday marks the minor low point of the week, dipping slightly to around 580 conversions before recovering.

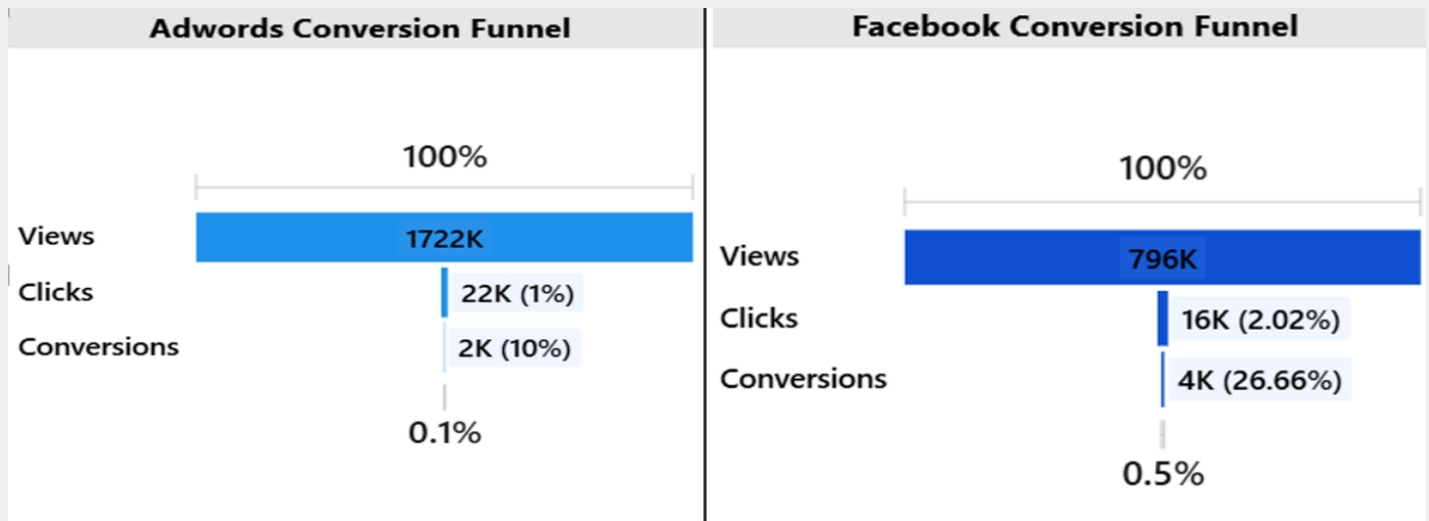
Monthly Facebook Trends:



- **Views:** Highly volatile throughout the year, peaking sharply in **August (~70.6K)** before dropping to an annual low in **October (~63K)**.
- **Clicks:** Shows an upward seasonal trajectory, recovering from a deep **February** low (~1,130) to reach its absolute peak in **October (~1,560)**.
- **Conversions:** Follows the click pattern tightly, hitting its lowest point in **February (~302)** and surging to an annual high in **October (>405)**.

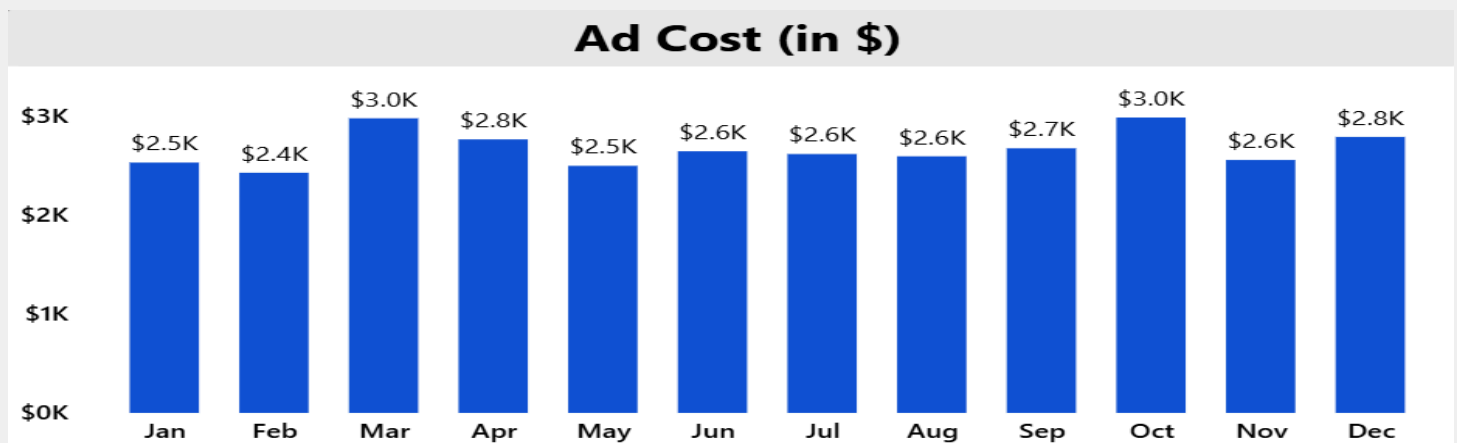
- **Executive Value:** Despite lower ad views in October, clicks and conversions annual highs during that month, highlighting massive campaign efficiency and a high conversion rate in Q4.
- **Adwords :** On the other hand google adwords generates more views and clicks but half the conversions as facebook

Campaign Funnel Comparison:



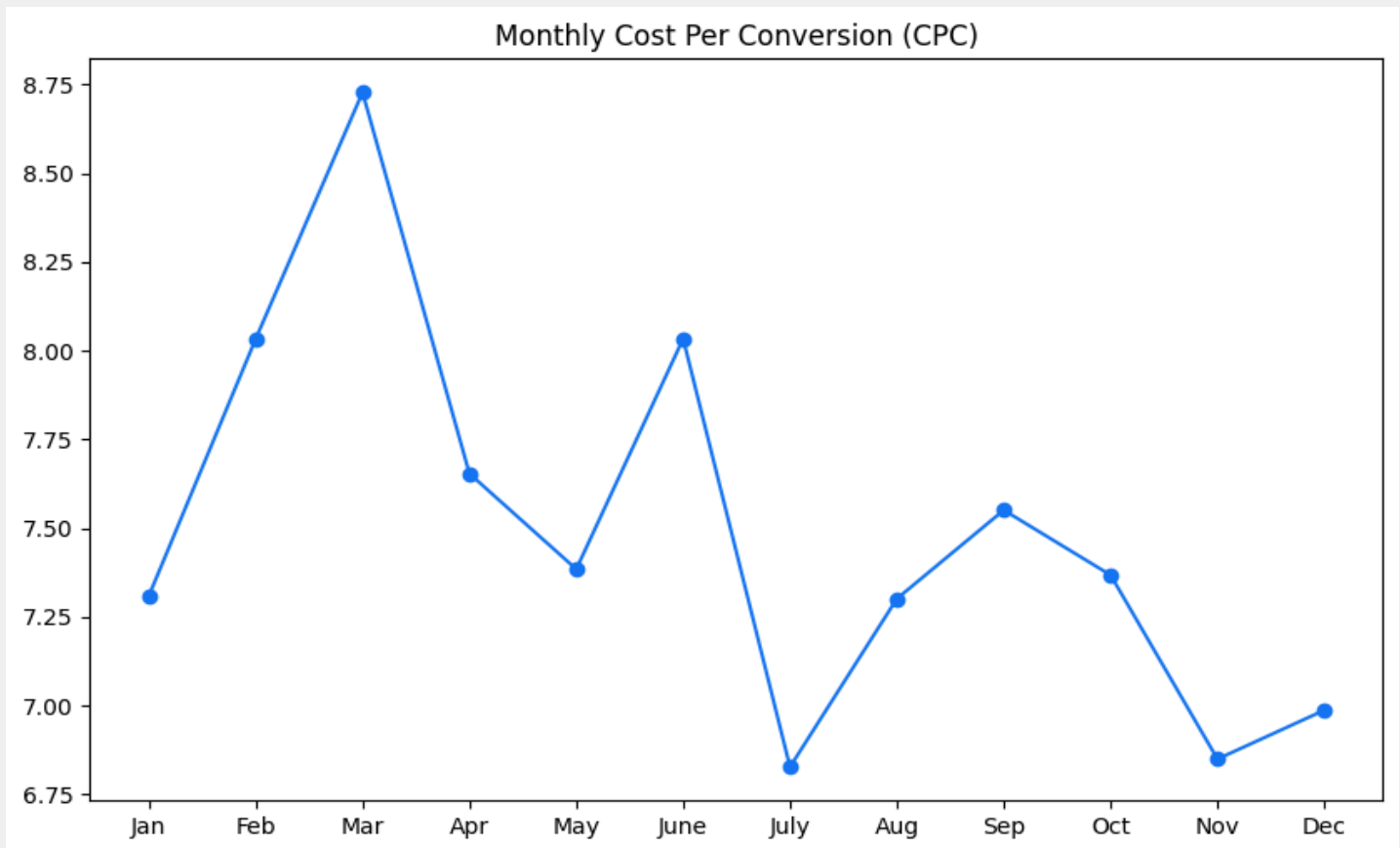
- **AdWords Funnel:** Drives massive exposure with **1,722K views**, but suffers from severe drop-offs, resulting in a low **1% click-through rate (CTR)** and a **10% click-to-conversion rate**, yielding an overall conversion efficiency of just **0.1%**.
- **Facebook Funnel:** Generates far fewer views (**796K**) but demonstrates vastly superior audience engagement, securing a **2.02% CTR** and a remarkable **26.66% click-to-conversion rate**.
- **Strategic Takeaway:** Facebook is **5x more efficient** overall than AdWords (0.5% vs. 0.1% total conversion rate), generating double the actual conversions (**4K vs. 2K**) from less than half the top-of-funnel exposure.

Monthly Ad Cost:



- **Budget Baseline:** Ad costs remain tightly managed and relatively stable, generally hovering between \$2.5K and \$2.8K per month for most of the fiscal period.
- **Spending Peaks:** Investment reaches its highest levels in March and October, with both months hitting an annual maximum spend of \$3.0K.
- **Spending Low:** Budget utilization drops to its lowest level in February, recording a minimal spend of \$2.4K.

Monthly Average Cost Per Conversion:



- **Overall Trajectory:** Despite prominent fluctuations in the first half of the year, the trend shows an overall negative slope, confirming a steady improvement in cost-efficiency by Q4.
- **Peak Acquisition Costs:** Costs peaked sharply in **March (~\$8.73)**, identifying it as the most expensive, highest-friction period for conversions.
- **Maximum Efficiency Benchmarks:** The campaign achieved its lowest and most profitable CPC levels in **July (~\$6.83)** and **November (~\$6.85)**.
- **Strategic Takeaway:** The final third of the year maintains a consistently lower cost baseline compared to the spring peaks, proving successful long-term campaign optimization.

Relationship between Advertising spend and Conversion rates

What this test means - the cointegration test checks if a stable, long-term relationship exists between Facebook Advertising Spend and Facebook Conversions. It determines if they move together reliably over time, rather than just fluctuating randomly.

- Hypotheses:
 - Null (**H₀**): There is no stable, long-term relationship between Facebook ad spend and conversions (they move randomly).
 - Alternative (**H₁**): A stable, long-term relationship exists between spend and conversions (they lock and move together over time).

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Cointegration test score: -14.755428385103219
P-value: 2.1337375979061323e-26

p-value is less than significance value, Reject the null hypothesis
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- **Meaning of the Numbers:**
 - Test Score (**-14.76**): Measures how tightly the two metrics are bound. Being exceptionally deep below zero proves a powerful, permanent statistical connection.
 - P-Value (2.13×10^{-26}): Measures the probability that this connection is just luck. Because it is effectively 0%, random chance is completely ruled out.
- **Stats Conclusion :**
 - What it means: ***This test proves that Facebook advertising isn't erratic; our budget and your conversion results are mathematically locked in a reliable, long-term partnership.***
 - Strategic Value: ***We can scale the Facebook ad budget up or down with absolute confidence, knowing with mathematical certainty that the conversion numbers will reliably follow the spend.***

Now with guarantee we can conclude that Facebook is the better and most efficient platform for campaigning, fetching us better ROI compared to Adwords.

Strategic Recommendations:

Based on the insights, statistical and market evidence collected across both campaigns, here are the actionable recommendations for the executive report to maximize client ROI and optimize resource allocation.

1. Budget Reallocation

- **Action :** Invert the current funding structure. Transferring at least 30% to 40% of the existing Google AdWords budget directly into the Facebook campaign ecosystem.

- **Reason :** AdWords currently consumes 60.59% (\$49K) of the budget to yield only 33.75% (2K) of total conversions. Facebook generates double the conversions (4K) using under 40% (\$32K) of the aggregate spend. This shift capitalizes on Facebook's 5x higher overall funnel efficiency (0.5% vs 0.1%).

2. Confident Scaling via Predictive Modeling

- **Action :** Deploy the validated linear regression model to scale Facebook budgets up safely, targeting a baseline threshold of 80 clicks per day to reliably forecast and hit a target of 19.31 daily conversions.
- **Reason :** The cointegration test score of -14.76 and a near-zero p-value mathematically prove that Facebook spend and conversions are locked in a permanent, non-random relationship. You can aggressively scale ad spend knowing with statistical certainty that conversions will reliably follow, eliminating typical ad-waste risk.

3. Capitalize on Q3 & Q4 and High-Efficiency Windows

- **Action :** Restructure the annual budget distribution to heavily weight spending toward the high-performing second half of the year (July, and September through December), with peak aggressive spending deployed in October. Conversely, pull back budgets during the high-friction, high-cost periods of February and March.
- **Reason :** October represents maximum campaign efficiency, hitting annual highs in clicks and conversions despite lower raw top-of-funnel views. Furthermore, Q3 & Q4 capture the lowest acquisition costs of the year, dipping to highly profitable benchmarks in July (\$6.83 CPC) and November (\$6.85 CPC) compared to the expensive March peak (\$8.73 CPC).

4. Implement Bidding Strategies for specific Day of week

- **Action :** Deploy an automated day-of-week bid multiplier strategy on Facebook. Programmatically increase ad bids on Sundays and Mondays to maximize impression share, and apply a minor bid markdown on Wednesdays.
- **Reason :** While weekly baseline performance is highly stable (~600 conversions/day), Monday serves as a distinct peak performance day (~660 conversions), whereas Wednesday represents the natural weekly low (~580 conversions).

5. Pivot AdWords Strategy to Inbound Intent (Intent vs. Discovery)

- **Strategic Context:** While Facebook out-converted AdWords in this parallel test, the two platforms capture fundamentally different buying behaviors. Facebook excels at **Passive Discovery** (using visual appeal like videos & photos to seek immediate attention and create brand demand), while Google Ads excels at capturing **Active Intent** (targeting users who are actively searching for an immediate solution they are facing problem with).

- **Action:** Do not abandon Google AdWords; instead, combine both platforms into a unified funnel. Transition AdWords away from broad view-generation to capture the awareness built by Facebook later, restricting its scope strictly to high-purchase-intent keywords while auditing the landing page to fix mid-funnel drop-offs. In short, use Facebook to build brand awareness (by marketing the product or service), and Google Ads to capture those users when they finally search for the product later.
- **Reason :** AdWords drove massive top-of-funnel exposure (~**2M views**) but suffered a devastating drop-off, registering a low **1% CTR** and a weak **0.1 conversions**. Restricting AdWords to pure high-intent search terms will eliminate wasteful spend on passive viewers while preserving the agency's ability to capture ready-to-buy clients.